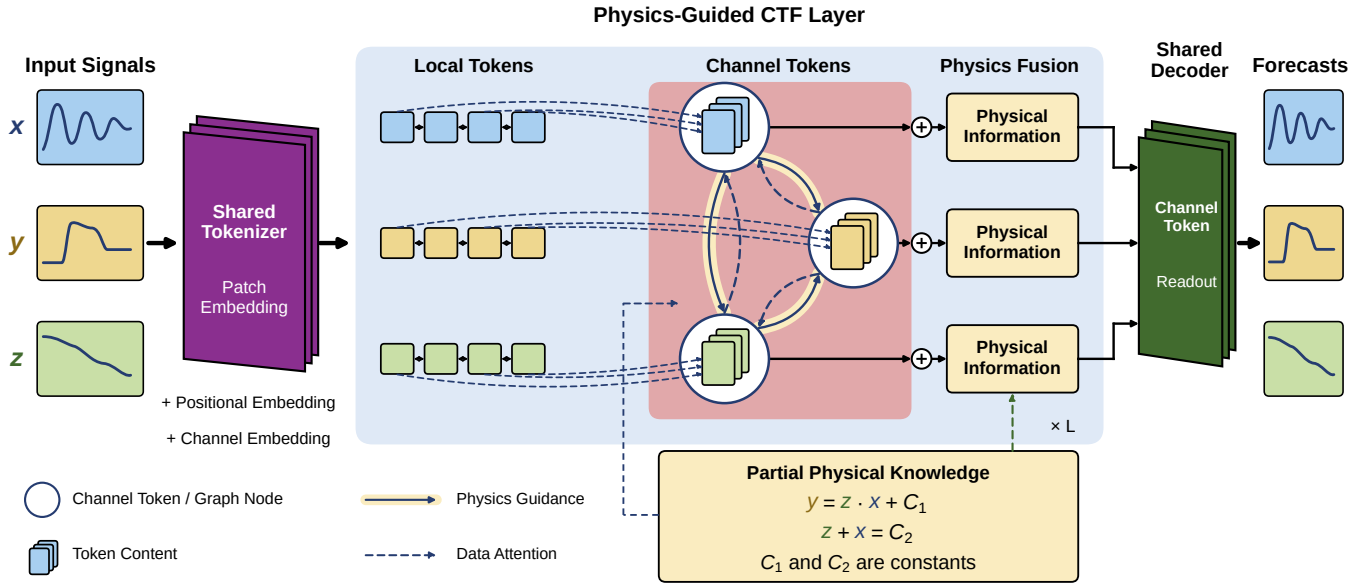
Physics-Guided CTF Layer
Input Signals
x
y
z
Shared Tokenizer
Patch Embedding
+ Positional Embedding
+ Channel Embedding
Local Tokens
Channel Tokens
Physics Fusion
Physical Information
Physical Information
Physical Information
× L
Shared Decoder
Channel Token
Readout
Forecasts
Partial Physical Knowledge
$y = z \cdot x + C_1$
$z + x = C_2$
C_1 and C_2 are constants
Channel Token / Graph Node
Token Content
Physics Guidance
Data Attention